theory

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1 Price Mortality Theory

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1.0.1 Overview

This script constructs a **data-driven volatility forecasting framework** built entirely from price data. Starting from raw log returns $r_t = \ln(P_t/P_{t-1})$, it estimates realized volatility $\hat{\sigma}_h(t)$ across multiple horizons, then builds a momentum surface $\mu(W, t)$ by differentiating the SMA field with respect to both window size W and time t. The core question is whether $\mu(W, t)$ — a scale-invariant measure of how aggressively the trend is changing — carries **predictive signal** for future volatility. This is tested non-parametrically via Spearman rank correlation across the full (W, h) grid, yielding a heatmap that reveals which trend-window and forecast-horizon combinations are most informative. The pipeline concludes by selecting the optimal pair (W^*, h^*) that maximises ρ_S, providing a principled, data-driven basis for choosing a forecasting configuration.

1.1 Environment Setup

1.1.1 Install Dependencies

[27]:
```
%cd /home/jose/signal-analysis
%pip install -r requirements.txt -q
%cd notebooks
```

```
/home/jose/signal-analysis
```

```
/home/jose/signal-analysis/venv/lib/python3.10/site-
packages/IPython/core/magics/osm.py:417: UserWarning: This is now an optional
IPython functionality, setting dhist requires you to install the `pickleshare`
library.
  self.shell.db['dhist'] = compress_dhist(dhist)[-100:]
```

```
Note: you may need to restart the kernel to use updated packages.
/home/jose/signal-analysis/notebooks
```

1.1.2 Import Libraries

[28]:
```
import yfinance as yf
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
from scipy.stats import spearmanr
```

1.1.3 Load Data

[29]:
```
ticker = "AAPL"

df = yf.download(ticker, period="5y", interval="1d", auto_adjust=True)
df["Day"] = np.arange(len(df))
df["Log Returns"] = np.log(df["Close"] / df["Close"].shift(1))
df.head()
```

```
[*********************100%***********************]  1 of 1 completed
```

[29]:
```
Price            Close        High         Low        Open     Volume Day  \
Ticker            AAPL        AAPL        AAPL        AAPL       AAPL
Date
2021-03-03  118.885887  122.440972  118.671607  121.564375  112966300   0
2021-03-04  117.006081  120.385846  115.535353  118.583956  178155000   1
2021-03-05  118.262550  118.769032  114.512669  117.833997  153766600   2
2021-03-08  113.334137  117.853476  113.188036  117.785297  154376600   3
2021-03-09  117.941101  118.885878  115.700916  115.934673  129525800   4

Price      Log Returns
Ticker
Date
2021-03-03         NaN
2021-03-04   -0.015938
2021-03-05    0.010681
2021-03-08   -0.042567
2021-03-09    0.039845
```

1.2 Example of Realized Volatility by Horizon (RVH) and Price Mortality (PM)

1.2.1 Realized Volatility by Horizon

For a time series of log returns $r_t = \ln(P_t/P_{t-1})$, the **realized volatility** over a rolling window of h periods is estimated as the sample standard deviation:

$$\hat{\sigma}_h(t) = \sqrt{\frac{1}{h-1}\sum_{i=0}^{h-1}\left(r_{t-i} - \bar{r}_{t,h}\right)^2}$$

where $\bar{r}_{t,h} = \frac{1}{h}\sum_{i=0}^{h-1} r_{t-i}$ is the rolling mean over the same window.

The code computes $\hat{\sigma}_h(t)$ for four horizons:

$$h \in \{5,\ 21,\ 63,\ 252\} \quad \text{(approx. 1 week, 1 month, 1 quarter, 1 year)}$$

Each estimate belongs to the family of **historical volatility** estimators. Under the assumption that returns are i.i.d. with variance σ^2, the estimator is unbiased:

$$\mathbb{E}\left[\hat{\sigma}_h^2\right] = \sigma^2$$

and its standard error shrinks with window size:

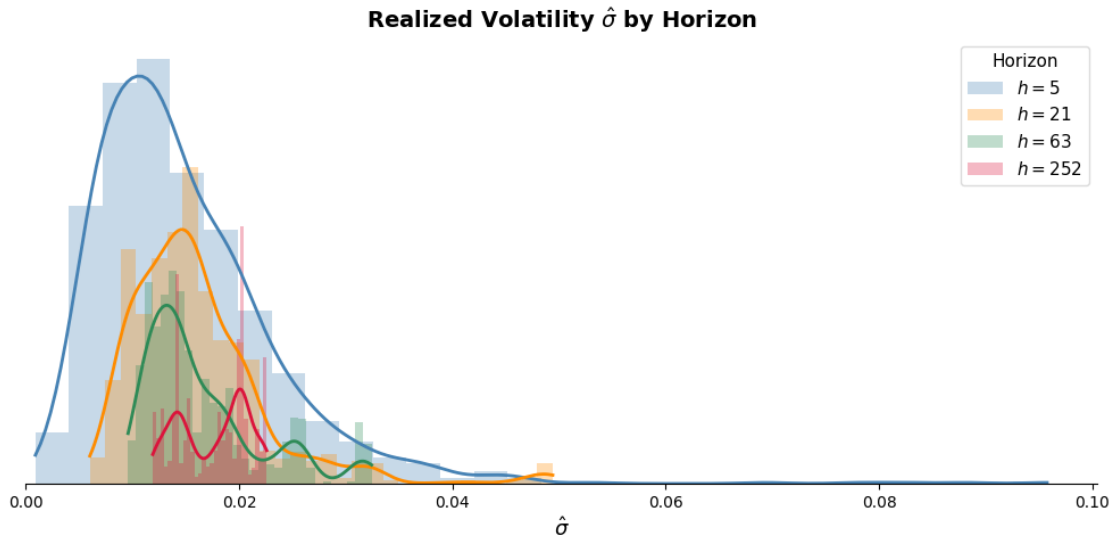
$$\mathrm{SE}(\hat{\sigma}_h) \approx \frac{\sigma}{\sqrt{2(h-1)}}$$

This means **longer horizons produce smoother, lower-variance estimates** of volatility, while **shorter horizons capture higher-frequency fluctuations** — a trade-off visible in the distribution plots, where the $h = 5$ distribution is wide and right-skewed, and the $h = 252$ distribution is narrow and more concentrated.

The distributions are plotted as **kernel density estimates (KDE)** overlaid on histograms, illustrating how the shape of $\hat{\sigma}_h$ evolves across horizons. Since $\hat{\sigma}_h \geq 0$ by construction, the x-axis is left-bounded at zero.

[30]:
```
h_values = [5, 21, 63, 252]
colors = ["steelblue", "darkorange", "seagreen", "crimson"]

fig, ax = plt.subplots(figsize=(10, 5))

for h, color in zip(h_values, colors):
    df[f"RV_{h}"] = df["Log Returns"].rolling(window=h).std()
    sns.histplot(
        data=df[f"RV_{h}"].dropna(),
        kde=True,
        bins=30,
        ax=ax,
        label=f"$h = {h}$",
        color=color,
        alpha=0.3,
        line_kws={"linewidth": 2},
        edgecolor="none",
    )

ax.set_xlim(left=0)
ax.set_xlabel(r"$\hat{\sigma}$", fontsize=13)
ax.set_ylabel("")
ax.yaxis.set_visible(False)
```

```
ax.spines["left"].set_visible(False)
ax.set_title(r"Realized Volatility $\hat{\sigma}$ by Horizon", fontsize=14,␣
↪fontweight="bold")
ax.legend(title="Horizon", fontsize=11, title_fontsize=11, framealpha=0.7)
ax.spines[["top", "right"]].set_visible(False)

plt.tight_layout()
plt.show()
```



1.2.2 Simple Moving Averages (SMA) & the Momentum Surface $\mu(W, t)$

1. Simple Moving Average For a price series P_t, the **Simple Moving Average** over a window of W periods is:

$$\mathrm{SMA}_W(t) = \frac{1}{W} \sum_{i=0}^{W-1} P_{t-i}$$

The code computes this for four horizons:

$$W \in \{5,\ 21,\ 63,\ 252\} \quad \text{(approx. 1 week, 1 month, 1 quarter, 1 year)}$$

Longer windows produce smoother estimates that suppress short-term noise, at the cost of increased **lag** — the SMA reacts more slowly to recent price changes. Formally, the lag of an SMA is:

$$\mathrm{lag}(W) = \frac{W-1}{2} \quad \text{periods}$$

---

2. The Momentum Surface $\mu(W, t)$ The 3D surface generalises the SMA into a **continuous field** over both window size W and time t, and defines a normalised momentum measure:

$$\mu(W,\, t) = -\frac{1}{\mathrm{SMA}} \left[\frac{\partial\, \mathrm{SMA}}{\partial W} + \frac{\partial\, \mathrm{SMA}}{\partial t} \right]$$

Each partial derivative captures a distinct effect:

Rate of change with respect to window size W:

$$\frac{\partial\, \mathrm{SMA}}{\partial W} \approx \frac{\mathrm{SMA}_{W+\Delta W}(t) - \mathrm{SMA}_{W-\Delta W}(t)}{2\, \Delta W}$$

This measures how sensitive the average is to the choice of smoothing horizon — i.e., how much the signal changes as we zoom out.

Rate of change with respect to time t:

$$\frac{\partial\, \mathrm{SMA}}{\partial t} \approx \frac{\mathrm{SMA}_W(t+1) - \mathrm{SMA}_W(t-1)}{2}$$

This is the temporal momentum of the moving average — positive when the trend is rising, negative when falling.

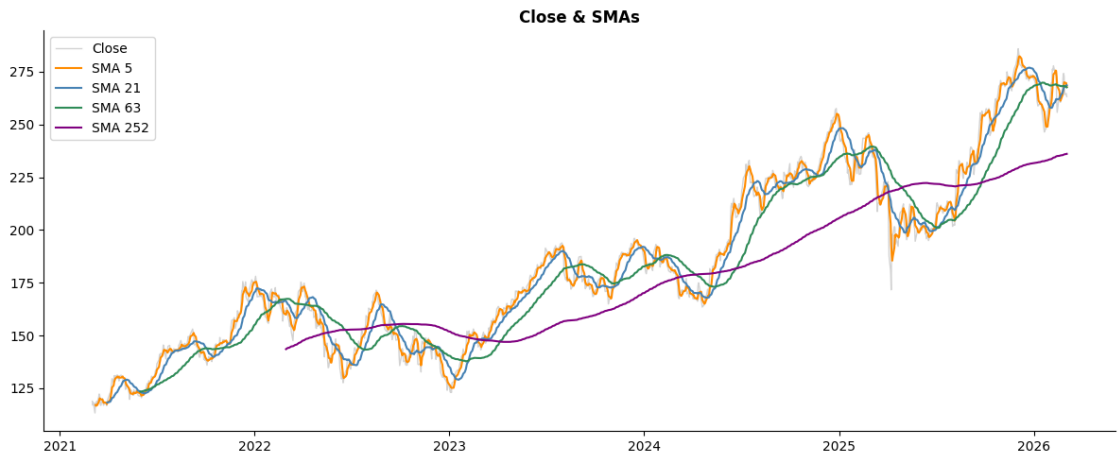
Normalisation by SMA makes μ scale-invariant (analogous to a log-derivative):

$$\mu(W, t) \approx -\frac{d}{d\cdot} \ln(\mathrm{SMA}(W, t))$$

so that μ can be compared across assets and time periods regardless of price level.

---

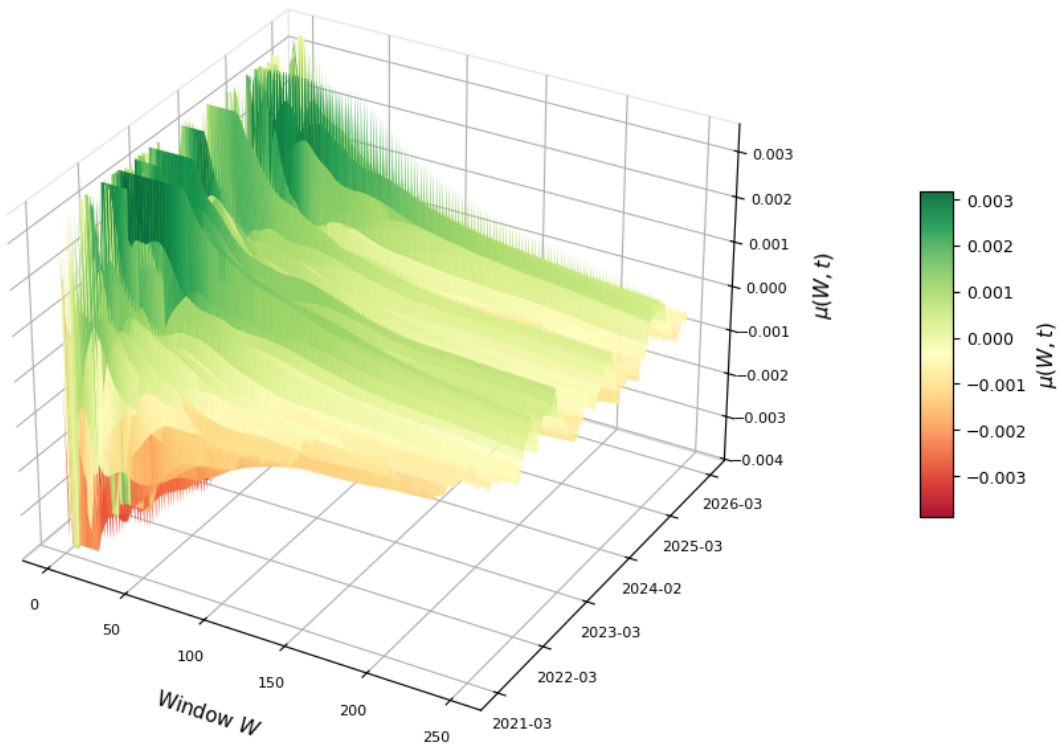
[31]:
```
window_arr = [5, 21, 63, 252]
colors = ["darkorange", "steelblue", "seagreen", "purple"]

fig, ax = plt.subplots(figsize=(12, 5))

ax.plot(df.index, df["Close"], color="lightgrey", linewidth=1, label="Close",␣
↪zorder=1)

for W, color in zip(window_arr, colors):
    ax.plot(df.index, df["Close"].rolling(W).mean(), color=color, linewidth=1.
↪5, label=f"SMA {W}")

ax.set_title("Close & SMAs", fontweight="bold")
ax.set_xlabel("")
ax.legend()
ax.spines[["top", "right"]].set_visible(False)
plt.tight_layout()
plt.show()
```



[32]:
```
fig = plt.figure(figsize=(14, 8))
ax = fig.add_subplot(111, projection="3d")

# Downsample for smoother rendering
step_w, step_t = 2, 3
W_ds = inner_windows[::step_w]
t_ds = np.arange(len(df))[::step_t]
mu_ds = mu[::step_t, ::step_w]
mu_ds = np.clip(mu_ds, np.nanpercentile(mu_ds, 1), np.nanpercentile(mu_ds, 99))

W_grid, t_grid = np.meshgrid(W_ds, t_ds)

surf = ax.plot_surface(
    W_grid, t_grid, mu_ds,
    cmap="RdYlGn",
    alpha=0.92,
    linewidth=0,
    antialiased=True,
    rcount=200,
    ccount=200,
)

# Colorbar
cbar = fig.colorbar(surf, ax=ax, shrink=0.4, aspect=10, pad=0.1)
cbar.set_label(r"$\mu(W, t)$", fontsize=11)
cbar.ax.tick_params(labelsize=9)

# Date ticks on time axis
n = len(df)
tick_idx = np.linspace(0, n - 1, 6, dtype=int)
tick_labels = [df.index[i].strftime("%Y-%m") for i in tick_idx]
```

```
ax.set_yticks(tick_idx)
ax.set_yticklabels(tick_labels, fontsize=8)

ax.set_xlabel("Window $W$", fontsize=11, labelpad=10)
ax.set_ylabel("", fontsize=11, labelpad=30)
ax.set_zlabel(r"$\mu(W, t)$", fontsize=11, labelpad=10)
ax.set_title(
    r"$\mu(W,t) =␣
↪-\frac{1}{\mathrm{SMA}}\left[\frac{\partial\,\mathrm{SMA}}{\partial W} +␣
↪\frac{\partial\,\mathrm{SMA}}{\partial t}\right]$",
    fontsize=13, pad=20
)

ax.view_init(elev=30, azim=-60)
ax.tick_params(axis="both", labelsize=8)
ax.xaxis.pane.fill = False
ax.yaxis.pane.fill = False
ax.zaxis.pane.fill = False
ax.xaxis.pane.set_edgecolor("lightgrey")
ax.yaxis.pane.set_edgecolor("lightgrey")
ax.zaxis.pane.set_edgecolor("lightgrey")
ax.grid(True, linestyle="--", linewidth=0.4, alpha=0.5)

plt.show()
```

$$\mu(W,t) = -\frac{1}{\text{SMA}}\left[\frac{\partial \text{SMA}}{\partial W} + \frac{\partial \text{SMA}}{\partial t}\right]$$



1.2.3 Spearman Correlation Between $\mu(W,t)$ and Forward Realized Volatility $\hat{\sigma}(t+h)$

1. Forward Realized Volatility For each forecast horizon h, the **forward realized volatility** is the rolling standard deviation of log returns computed h steps ahead:

$$\hat{\sigma}_h(t) = \sqrt{\frac{1}{h-1}\sum_{i=0}^{h-1}\left(r_{t+i} - \bar{r}_{t,h}\right)^2}$$

This is a **forward-looking** quantity — at time t, it summarises the volatility that will be *realised* over the next h periods, making it a natural target for forecasting.

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2. Spearman Rank Correlation For each pair (W,h), the code computes the **Spearman rank correlation** between the momentum signal $\mu(W,t)$ and the forward volatility $\hat{\sigma}_h(t)$:

$$\rho_S(W,h) = 1 - \frac{6\sum_t d_t^2}{n(n^2-1)}, \quad d_t = \text{rank}(\mu(W,t)) - \text{rank}(\hat{\sigma}_h(t))$$

Spearman's ρ is preferred over Pearson's here for two reasons:

- **Robustness** — financial returns and volatility are heavy-tailed; rank-based measures are insensitive to outliers
- **Monotonic sensitivity** — it captures any monotonic relationship between μ and $\hat{\sigma}$, not just linear ones

A minimum sample guard of $n \geq 30$ overlapping observations is enforced before computing ρ_S to avoid spurious correlations from small samples.

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3. The Correlation Matrix The result is a matrix $\mathbf{R} \in \mathbb{R}^{|W| \times |h|}$ where each entry is:

$$\mathbf{R}_{i,j} = \rho_S\Big(\mu(W_i, \cdot),\ \hat{\sigma}_{h_j}(\cdot)\Big), \quad W_i \in [2, 252], \quad h_j \in [1, 125]$$

The axes therefore span:

Axis	Variable	Range	Interpretation
x	Forecast horizon h	$1 \to 125$	How far ahead we predict volatility
y	SMA window W	$2 \to 252$	How long a trend signal we use

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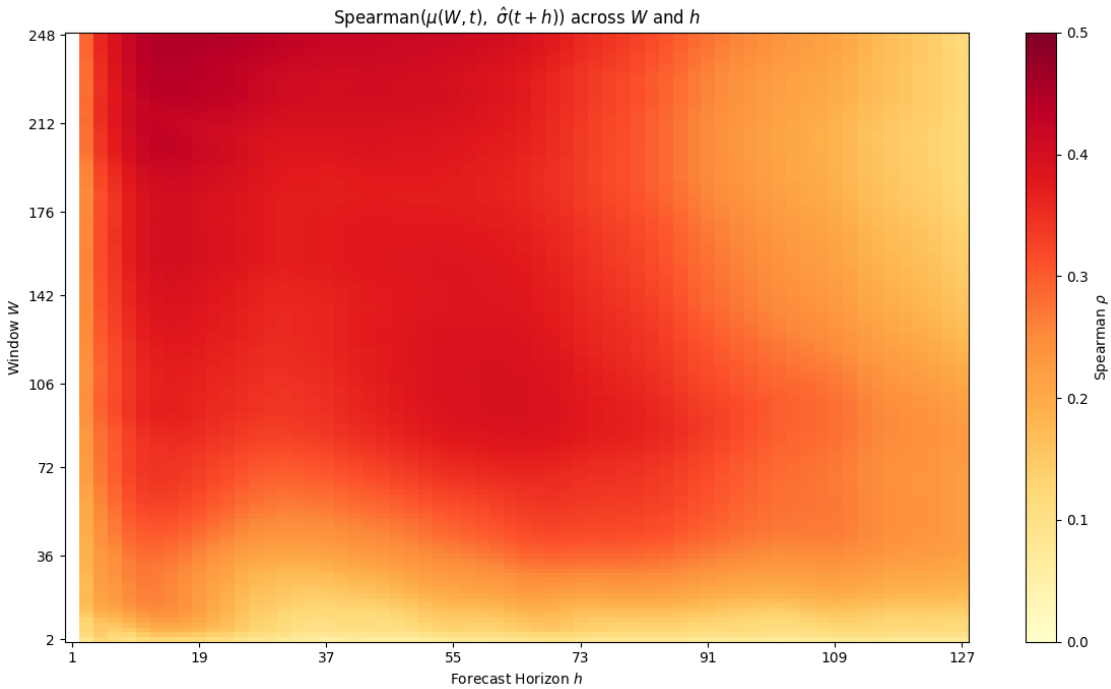
4. Interpretation A positive $\rho_S(W, h)$ indicates that **larger momentum μ today predicts larger realised volatility h steps ahead**, for a trend signal built on window W. Regions of the heatmap where ρ_S is high reveal which (W, h) combinations carry the most predictive information:

- **High ρ_S at small h** — short-horizon momentum predicts near-term volatility spikes
- **High ρ_S at large W** — longer-term trend signals are more persistent predictors
- **Diagonal structure** (if present) — suggests a natural alignment between the trend window and the forecast horizon, i.e. $W \sim h$

The colormap is bounded at $\rho_S \in [0, 0.5]$, reflecting the expectation that predictability is positive but modest — consistent with weak-form market efficiency.

[33]:
```
W_values = np.arange(2, max(inner_windows), 2)
h_values = np.arange(1, 126+2, 2)

# Forward RV surface
rv_surface = pd.concat(
    {h: df["Log Returns"].shift(-h).rolling(h).std() for h in h_values},
    axis=1
)
rv_surface.columns = h_values

# mu surface - already computed as mu (numpy array, rows=t, cols=W)
# rebuild as DataFrame for alignment
mu_surface = pd.DataFrame(mu, index=df.index, columns=inner_windows)
```

```python
mu_surface = mu_surface[W_values]  # coarsen to match W_values

# Correlation matrix: rows=W, cols=h
corr_matrix = np.full((len(W_values), len(h_values)), np.nan)

for i, W in enumerate(W_values):
    for j, h in enumerate(h_values):
        x = mu_surface[W]
        y = rv_surface[h]
        mask = x.notna() & y.notna()
        if mask.sum() > 30:  # minimum sample guard
            corr_matrix[i, j], _ = spearmanr(x[mask], y[mask])

# Plot as heatmap
fig, ax = plt.subplots(figsize=(12, 7))
im = ax.imshow(corr_matrix, aspect="auto", cmap="YlOrRd", vmin=0, vmax=0.5,␣
 ↪origin="lower")
fig.colorbar(im, ax=ax, label=r"Spearman $\rho$")

ax.set_xticks(np.linspace(0, len(h_values)-1, 8, dtype=int))
ax.set_xticklabels(h_values[np.linspace(0, len(h_values)-1, 8, dtype=int)])
ax.set_yticks(np.linspace(0, len(W_values)-1, 8, dtype=int))
ax.set_yticklabels(W_values[np.linspace(0, len(W_values)-1, 8, dtype=int)])

ax.set_xlabel("Forecast Horizon $h$")
ax.set_ylabel("Window $W$")
ax.set_title(r"Spearman$(\mu(W,t),\ \hat{\sigma}(t+h))$ across $W$ and $h$")
plt.tight_layout()
plt.show()
```



1.2.4 Optimal (W^*, h^*) Selection

The optimal pair is found by maximising the Spearman correlation over the full grid:

$$(W^*, h^*) = \underset{W,\,h}{\arg\max}\ \rho_S(\mu(W,t),\ \hat{\sigma}_h(t))$$

located via `np.nanargmax` on the correlation matrix $\mathbf{R}$, which identifies the index of the globally highest ρ_S while ignoring any (W, h) pairs that failed the minimum sample guard ($n < 30$).

The resulting W^* and h^* define the **most predictive configuration** — the trend-window and forecast-horizon combination for which today's momentum signal has the strongest monotonic relationship with future realised volatility.

[34]:
```
# Find optimal (W, h) pair
best_idx = np.unravel_index(np.nanargmax(corr_matrix), corr_matrix.shape)
best_W = W_values[best_idx[0]]
best_h = h_values[best_idx[1]]
best_rho = corr_matrix[best_idx]

print(f"Optimal Window:  W = {best_W} days")
print(f"Optimal Horizon: h = {best_h} days")
print(f"Spearman  :      {best_rho:.4f}")
```

```
Optimal Window:  W = 248 days
Optimal Horizon: h = 15 days
```

```
Spearman  :      0.4490
```